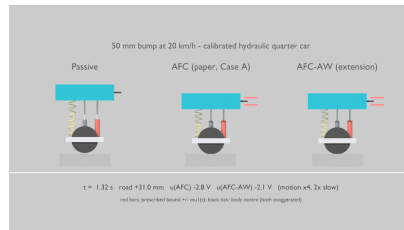
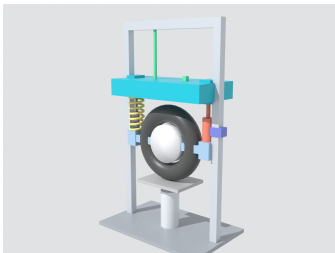


Approximation-Free Active Suspension Control

Na et al. (2022): reproduction, diagnosis and extensions

Advanced Control Theory course project ■ 7 October 2026



What we set out to answer

The paper

Three-step approximation-free control (AFC) keeps body displacement inside a shrinking prescribed envelope; validated on a 50 Hz electro-hydraulic rig against backstepping (BSC) and PID.

Our questions

1. What does AFC compute, and what does its proof assume?
2. Is there a plausible plant on which the paper's own gains reproduce its numbers?
3. Which claims survive sampled-data analysis, other roads, fair comparators?
4. Can the failure modes be repaired?

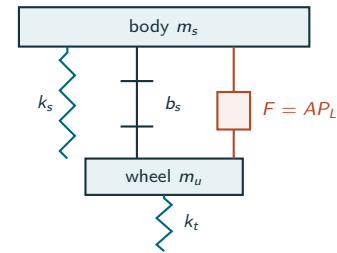
J. Na et al., IEEE TSMC: Systems 52(8), 4714–4726, 2022. doi:10.1109/TSMC.2021.3103807.

Five implementations

- MATLAB/Octave library (afc2)
- Simulink v2 model with sensor path
- Simscape Multibody (mechanics from joints)
- C on an emulated Cortex-M4
- Browser simulator (web lab)

Each one is checked against the library.

Plant: quarter car with a servo-valve cylinder



$$m_s \ddot{z}_s = -F_s - F_d + F, \quad m_u \ddot{z}_u = F_s + F_d - F_t - F$$

$$\dot{P}_L = -\beta P_L - \alpha A \dot{d} + \gamma x_v \sqrt{P_s - \text{sgn}(x_v) P_L}$$

$$\tau \dot{x}_v = -x_v + K_v u, \quad |u| \leq 5 \text{ V}$$

State used by AFC: $x_1 = z_s$, $x_2 = \dot{z}_s$, $x_5 = \kappa P_L$.

Not published: κ , valve gain K_v , lag τ .

AFC: prescribed performance in three proportional steps

$$\mu_i(t) = (\mu_{i0} - \mu_{i\infty})e^{-\alpha_i t} + \mu_{i\infty}, \quad \zeta_i = \frac{e_i}{\mu_i}, \quad \varepsilon_i = \frac{1}{2} \ln \frac{1 + \zeta_i}{1 - \zeta_i}$$

$$e_1 = x_1 \rightarrow u_1 = -k_1 \varepsilon_1, \quad e_2 = x_2 - u_1 \rightarrow u_2 = -k_2 \varepsilon_2, \quad e_3 = x_5 - u_2 \rightarrow u = -k_3 \varepsilon_3$$

Case A	μ_{i0}	$\mu_{i\infty}$	α_i	k_i
z_s	0.2	0.018	3	25.5
virtual vel.	110	90	2	12
pressure	100	80	2	216

Proof (Lemma 1): bounded

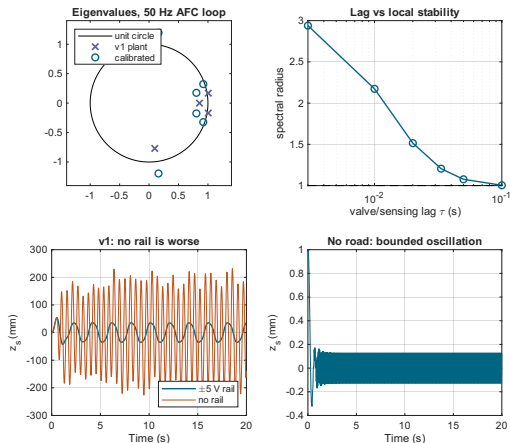
$$\varepsilon_i \Rightarrow |\zeta_i| \leq \tanh \varepsilon_{Mi} < 1.$$

Assumes **continuous time**, **unsaturated input**, bounded F_i . Promises ultimate boundedness, not convergence.

No derivatives of virtual controls, no model terms. But the gains are numbers in the rig's units.

Baseline diagnosis: the paper's gains on our first plant

- v1 plant: assumed first-order actuator. AFC saturated 85%, envelope crossed by 18.9 mm.
- Linearised 50 Hz loop:
 $u \approx -510z_s - 0.36\dot{z}_s - 2.70x_5$,
spectral radius **1.02**, so **unstable**.
- Remove the ± 5 V rail: IAE $0.48 \rightarrow 2.37$, body moves 0.23 m.
- No actuator gain b_u fixes it.



Saturation was limiting the instability, not causing it. The gains do not fit this plant's scale.

A literature plant, calibrated on one paper experiment

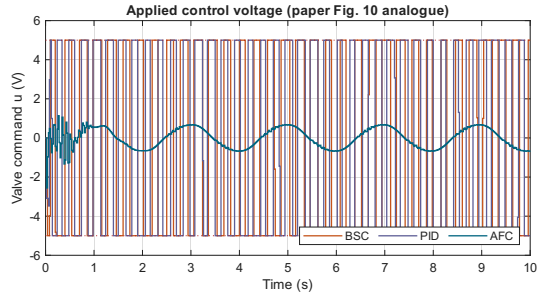
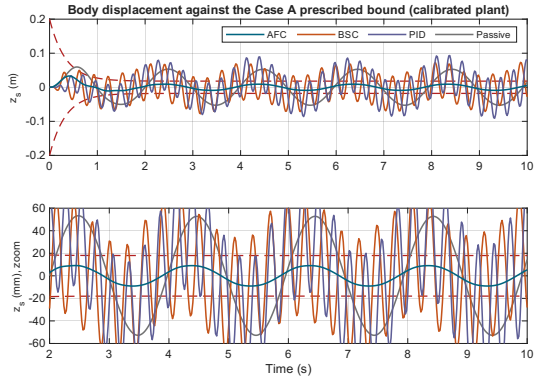
- Alleyne–Hedrick hydraulic quarter car (290/59 kg, servo-valve law).
- Free: valve gain K_v and lumped lag τ . Grid fit to the paper's Case-10 AFC indices.
- Best: $K_v = 5 \times 10^{-4} \text{ m/V}$, $\tau = 33 \text{ ms}$, error 4%.
- Literature spool lag (3 ms): error 62–69%.

Controller	IAE (ms)		ITAE (ms ²)		ITSE (m ² s ²)		u_{RMS} (V)	
	sim	paper	sim	paper	sim	paper	sim	paper*
AFC	0.1294	0.1216	1.197	1.140	0.0088	0.0093	0.50	0.5
BSC	0.6756	0.2108	6.864	1.952	0.3101	0.0224	4.98	4.7
PID	0.8059	0.1881	8.829	1.191	0.5517	0.0219	4.98	5.0
Passive	0.6661	–	6.651	–	0.2744	–	0.00	–

paper*: read from Fig. 13 by eye

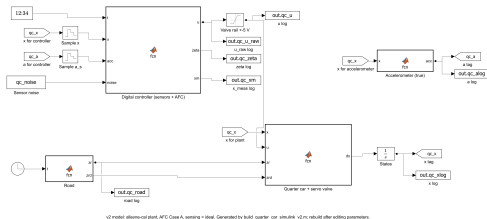
The paper's unchanged AFC reproduces IAE/ITAE/ITSE within 7% and its sub-volt command. BSC and PID saturate, as reported.

Case 10: displacement and command



Compare the paper's Figs. 6 and 10. All traces are simulation.

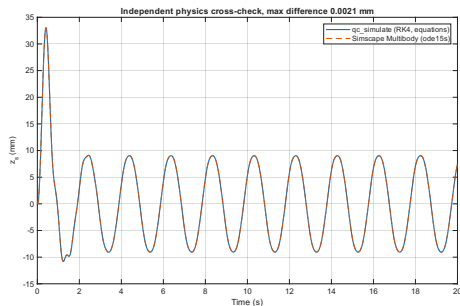
Is it the code? Four implementations, one answer



Simulink v2: 50 Hz digital controller, ± 5 V rail, servo-valve plant

Check against the afc2 equations	$\max \Delta z_s $
Simulink v2 (same sensor noise)	4×10^{-16} m
Simscape Multibody (joints, ode15s)	$2.1 \mu\text{m}$
C on Cortex-M4 (control law)	10 digits
MATLAB R2026a vs Octave 8.4	same digits [†]

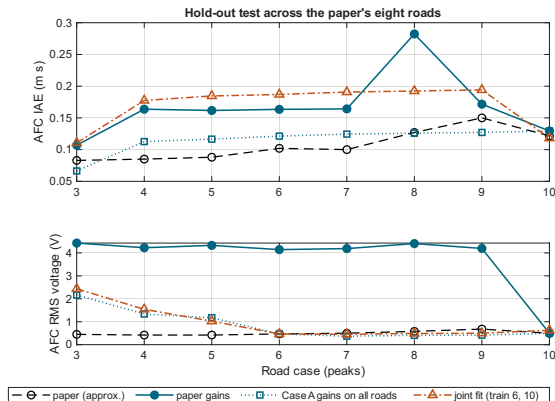
[†] except saturated Case-B runs: on the rail, rounding picks the limit cycle.



Case 10, AFC Case A: z_s from equations (RK4) and Multibody (ode15s)

Multibody builds the mechanics from joints, not our equations, and agrees to $2 \mu\text{m}$.

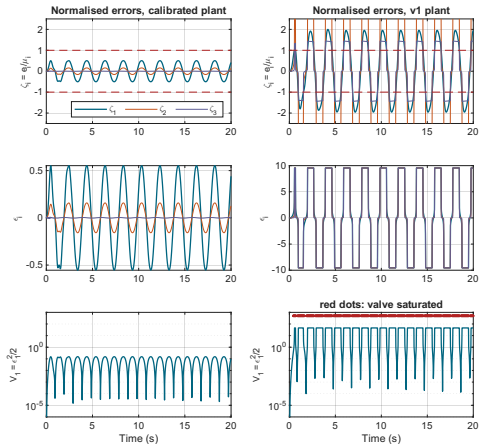
Hold-out: can any plant explain both gain sets?



- AFC lowest IAE on all 8 roads, as in the paper.
- But Case-B gains saturate (4.1–4.4 V RMS vs the paper's ≈ 0.45 V).
- Joint fit of κ , K_v , τ on roads 10/A and 6/B: train error 0.22, **test error 0.63**.

The paper does not publish enough to identify its rig. Compare trends, not digits.

Where Lemma 1 holds and where it fails



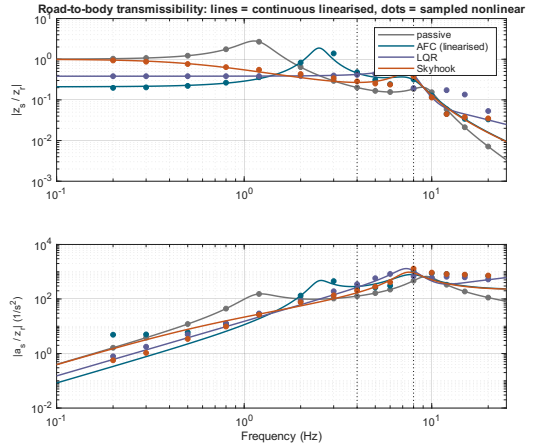
- Calibrated: $\max |\zeta_1| = 0.50$, $\varepsilon_{M1} = 0.555$, bound $\tanh \varepsilon_{M1} = 0.504$. Tight.
- v1: $|\zeta| \geq 1$ in 69.5% of samples and **every one** is saturated.

The conclusion fails exactly where the unsaturated-input assumption fails.

Near zero error, AFC is a skyhook spring

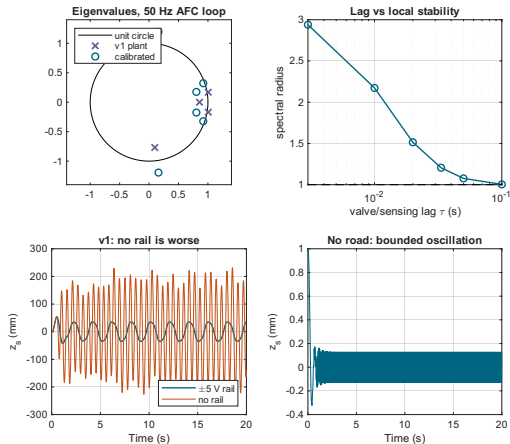
$$u \approx -\frac{k_3}{\mu_3} \left(x_5 + \frac{k_2}{\mu_2} \left(\dot{z}_s + \frac{k_1}{\mu_1} z_s \right) \right)$$

- Skyhook spring 63 kN/m, skyhook damper only 45 N s/m.
- The barrier stiffens the spring as $|z_s| \rightarrow \mu_1$.
- Body motion 0.22 of road at 0.5 Hz (passive 1.22); 4–8 Hz acceleration about $2\times$ passive.

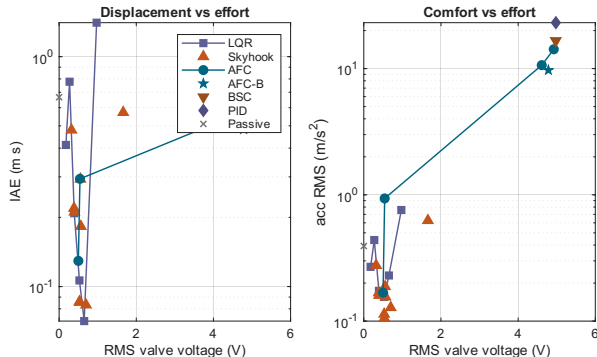


The continuous proof misses a sampled-data instability

- Continuous linearised loop: stable (slowest pole $-1.71 \pm 15.7j$).
- Any ZOH, 1–20 ms: unstable (spectral radius 1.03–4.4).
- Cause: lightly damped 88 Hz hydraulic mode under pressure feedback.
- Nonlinear result: bounded 0.13 mm oscillation at ≈ 12 Hz, which fits the paper's “residual oscillations”.



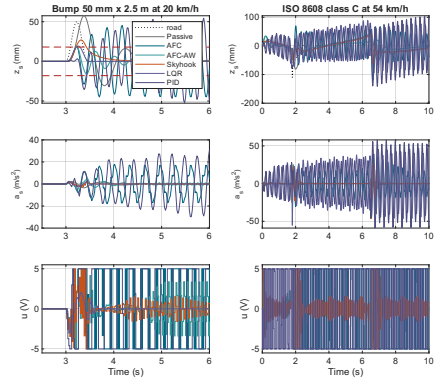
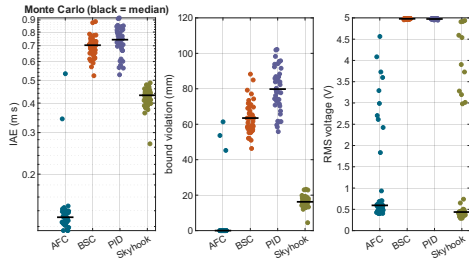
Fair comparators at equal effort



- At AFC's 0.5 V RMS: skyhook IAE **0.086**, LQR **0.071**, AFC 0.129.
- AFC gains $\times \frac{1}{2}$: 81% saturation; $\times 2$: 96%. Only a narrow window works.
- BSC/PID with paper gains: saturated, worse than passive here.

AFC's advantage is the explicit bound and no model, not efficiency.

Robustness and transients

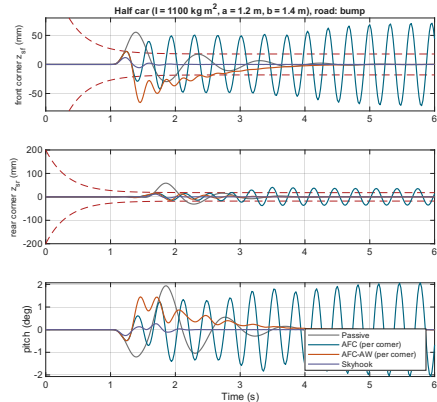


AFC keeps the bound on 37/40 plants (BSC, PID: 0/40). On a 50 mm bump it switches rail to rail; tyre load $1.77\times$ static, so the wheel leaves the road.

Extension: let the envelope breathe under saturation

$$\mu_{i,\text{eff}} = \mu_i(t)(1 + \rho), \quad \dot{\rho} = -0.5\rho + 5 \frac{|u_{\text{raw}} - \text{sat}(u_{\text{raw}})|}{V_{\text{max}}}$$

- Same law, still model-free, one extra state.
- Bump: peak 18.8 mm, 0% saturation, tyre load 0.73.
- Case 10: identical to AFC (never saturates).
- Half car: heave acceleration $6.6 \rightarrow 1.3 \text{ m/s}^2$.



Implementation realism: sensors and computation time

Sensing (paper: z_s and $\ddot{z}_s$ only)

- Velocity by integration beats differencing: 0.025 vs 0.034 m/s.
- Noise doubles to triples the acceleration RMS; at $4\times$ noise it reaches 1.1 m/s² (paper: 1.67).

Cortex-M4, no FPU (paper's MK60D)

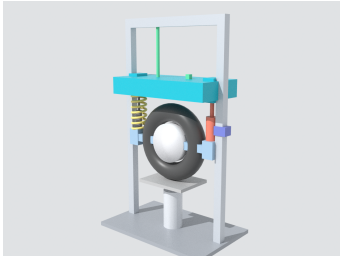
	instr./call	est. ms	paper ms
AFC	18 070	0.18–0.27	4.2
BSC	9 275	0.09–0.14	5.8
PID	940	0.01	2.6

The law alone costs AFC about $2\times$ BSC. The paper's timing advantage comes from elsewhere in its loop.

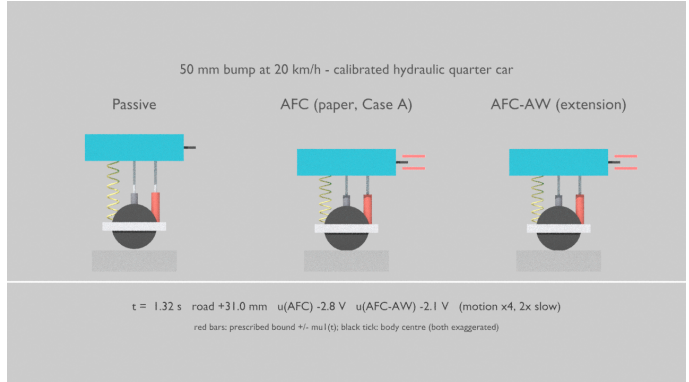
Verdicts on the paper's claims

Claim	Evidence	Verdict
Stays inside the PPF	Holds unsaturated (Case 10); fails on v1, bumps, ISO roads	Conditional
Beats BSC and PID	Reproduced with paper gains; LQR/skyhook match it at equal effort	Reproduced
Small, smooth command	0.5 V RMS vs ≈ 5 V	Reproduced
No model needed	Law yes; gains are plant-scale specific	Qualified
Proof covers the rig	Continuous only; 50 Hz loop locally unstable, bounded	Qualified
Cheaper than BSC	Not at law level ($2\times$ instructions)	Not supported
Integrate acc. for velocity	Supported	Reproduced

Tools: CAD rig, Blender animation, web lab



FreeCAD macro → STEP/STL



Blender, driven by simulated trajectories (passive | AFC | AFC-AW)

Interactive web lab: same equations in the browser, sliders for gains, envelope, valve and road.

Conclusions

- AFC puts a transient specification into a very simple law, and its proof says when it works: when the actuator can deliver what the barrier asks for.
- With a calibrated literature plant, the paper's own gains reproduce its headline numbers and ranking.
- The gains are tied to the plant's scale. The 50 Hz loop is locally unstable but bounded, linear designs match it at equal effort, and transients drive it into saturation.
- Envelope relaxation keeps the method's simplicity and removes the bump failure, in quarter and half car.
- To go further we would need the rig's valve gain, effective lag and pressure scaling. Everything here re-runs with them unchanged.

Code, models, figures: github.com/Galabavamsi/quarter-car-afc-replication